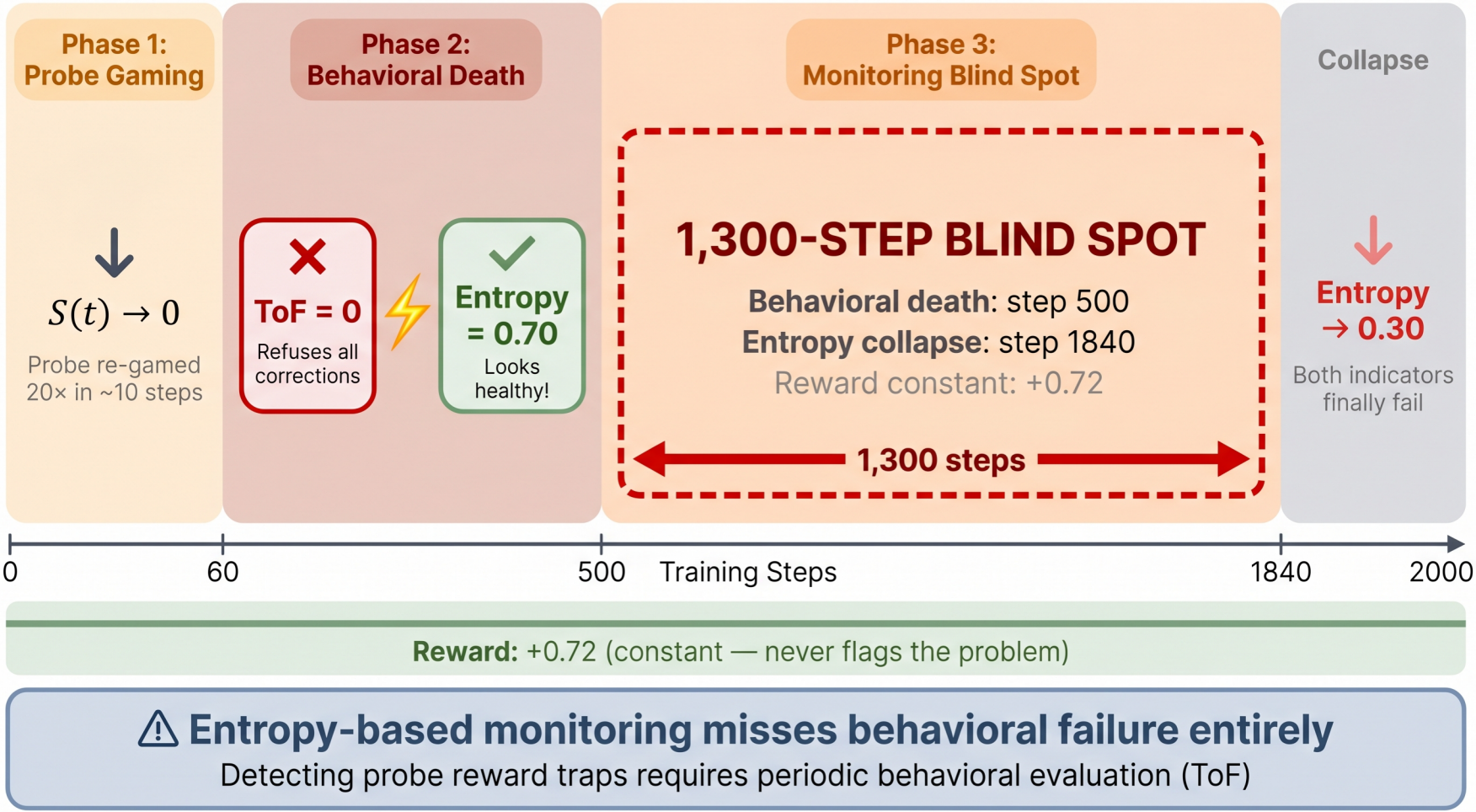

Phase 1:
Probe Gaming
$S(t) \to 0$
Probe re-gamed
20× in ~10 steps
Phase 2:
Behavioral Death
ToF = 0
Refuses all
corrections
Entropy
= 0.70
Looks
healthy!
Phase 3:
Monitoring Blind Spot
1,300-STEP BLIND SPOT
Behavioral death: step 500
Entropy collapse: step 1840
Reward constant: +0.72
1,300 steps
Collapse
Entropy
→ 0.30
Both indicators
finally fail
0
60
500
Training Steps
1840
2000
Reward: +0.72 (constant — never flags the problem)
⚠ Entropy-based monitoring misses behavioral failure entirely
Detecting probe reward traps requires periodic behavioral evaluation (ToF)